Department of Electrical & Electrical Engineering

BRAC University

Semester- Spring 2026

# EEE308L

**Electronic Circuits II**

**Section: 03**

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## Design Project :

**Group Members:**

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# Contribution:-

**Mahdi Zaman:**

- Design and Calculation.
- Simulation and troubleshooting.
- Trial and error process.
- Hardware Setup.
- Writing the report and creating data tables and comparison.

**Souvik Barman Ratul:**

- Simulation and Error calculation.
- Project Report writing.
- Collecting Results and arranging them.
- Formatting the project report.

**Faria Rahman Oyshi**

- Hardware Setup
- Hardware Implementation and Result Collection.
- Fine tuning the output and input graph.

## Objective:-

The aim of this project is to develop a digital thermometer by using a Wheatstone bridge circuit with an RTD sensor and an instrumentation amplifier. The system is designed so that the output voltage varies linearly from 0V at 0°C to 5V at 100°C, making temperature measurement clear. The instrumentation amplifier is used to amplify the small voltage produced by the bridge while ensuring that it does not draw significant current or disturb the circuit. This helps maintain accurate and stable readings. The system converts temperature changes into a corresponding voltage signal which can be easily measured. Overall, the project provides a clear understanding of how sensing elements and amplification circuits are combined in practical electronic systems.

## Design Methodology:-

The goal of this project is to build a digital thermometer that converts temperature into a voltage signal 0V to 5V. The system consists of two main parts: a Wheatstone Bridge for sensing and an Instrumentation Amplifier for signal scaling.

### 1.Wheatstone Stage:

The bridge uses an **RTD** (Resistance Temperature Detector) whose resistance changes based on the formula

$$R_{RTD} = R_O (1+\alpha T)$$

To make sure the output starts at 0V when the temperature is 0°C, the bridge is balanced by setting the other three resistors equal to the RTD's base resistance $R_O$

Here, At T = 0°C , Resistance = $R_O$

The bridge output is given as

$$v_{od} = \frac{\delta}{4} V_{REF}$$

Using $\alpha$= 0.002/°C and $V_{REF}$= 10V

- At **0°C** $\mathbf{v_{od}}$ = 0V
- At **100°C** $\mathbf{v_{od}}$ = 0.5V

## 2.Amplifier Stage:

We have two types of amplifiers to use in this case,

1. Difference Amplifier: This uses one op-amp. While simple, it has low input impedance. This would draw current from the bridge, causing a "loading effect" that makes the temperature readings inaccurate.

2. Instrumentation Amplifier: This uses three op-amps. It has extremely high input impedance, which satisfies the project requirement that the amplifier should not draw current from the bridge. It also has a high Common-Mode Rejection Ratio (CMRR), meaning it ignores the background DC voltage from the bridge and only amplifies the temperature difference.

After many considerations we went for an Instrumentation Amplifier.

## Reason behind using instrumentation amplifier

The Wheatstone bridge output voltage, $v_{od}$ is very small and delicate. To convert this signal into a usable 0V to 5V range for the thermometer without losing accuracy, an instrumentation amplifier is required. The specific reasons for choosing an instrumentation amplifier .

1. High Input Impedance

   The Wheatstone bridge is very sensitive. If we connect a normal amplifier, it will draw current from the bridge and change the voltage we are trying to measure. An instrumentation amplifier has "High Input Impedance" meaning it acts like a wall that sees the voltage but does not draw any current. This keeps the bridge reading accurate.

2. High Voltage Gain

   The maximum output from our bridge is only 0.5V but we need 5v for our thermometer scale. We need to increase the signal by 10 times. This amplifier is designed to provide this high gain very accurately without distorting the signal.

3. Single Resistor Tuning

   In many amplifiers, you have to change multiple resistors to change the gain, which is difficult to balance. In this circuit, we can set the exact gain we need (Gain = 10) by changing only one resistor . This makes the hardware setup much simpler.

4. Better Noise Rejection

   Since the signal is very small, it can be affected by noise. An instrumentation amplifier has good common-mode rejection capability, which helps in reducing noise and improving signal quality.

5. Software Implementation

   To validate the circuit theory, a complete model was first constructed in PSpice. Since a real-time RTD sensor cannot be directly interfaced in the software environment a variable resistor was substituted to represent the sensor’s behavior. The resistance was manually adjusted to simulate temperature variations. This allowed for the measurement of output voltages at various intervals between 0 ℃ and 100℃ confirming that the instrumentation amplifier correctly processes the bridge output.

6. Hardware Implementation

   After the simulation results are verified, the circuit is implemented on a breadboard using the required components. A potentiometer is used in place of the RTD to represent changes in temperature. Operational amplifiers are used to build the instrumentation amplifier circuit. The input and output voltages are measured using a multimeter for different resistance settings to observe the circuit behavior.

## Design and Analysis

Our goal is to vary the Vo from 0 V to 5V. In here, 0V output means 0 ℃ and 5V output means 100℃

The given equations ,

$$v_{od} = \frac{\delta}{4} V_{REF}$$

Where,

$\delta = \alpha T$

$\alpha = 0.002/℃$

$V_{REF} = 10V$

Now,

$$v_{od} = \frac{\delta}{4} V_{REF}$$

$$= \frac{0.002T}{4} * 10$$

$$= 0.005T$$

**For Instrumentation Amplifier:**

Input voltage range-

at,0℃, $\boldsymbol{v_{od}}$ = 0V to at,100℃, $\boldsymbol{v_{od}}$ = 0.5V

The input voltage will have to vary from 0 to 0.5 V. now for gain

$Av$ = (Ro/Ri)

= (5/0.5)

= 10

Where, $Av = \frac{R2}{R1} * (1 + \frac{2Rf}{RF})$

For ease of calculation, we took similar value for R1 and R2 @ 1kΩ,

This results into the Term $\frac{R2}{R1}$ cancelling out and we are left with,

$Av = (1 + \frac{2Rf}{RF})$

Here, $Av$ = 10

Or, $10 = (1 + \frac{2Rf}{RF})$

Or, $9 = \frac{2Rf}{RF}$

Or, $9RF = 2Rf$

From this relation we can synthesize an equation,

$$\boldsymbol{RF = \frac{2}{9}Rf}$$

## For Wheatstone Bridge:

Due to the laboratory environment's constraints, a physical RTD sensor was emulated using a precision variable resistor $R_T$. This allowed us to simulate specific thermal conditions by manually adjusting the resistance to match the theoretical values derived from the RTD temperature coefficient formula:

$$\mathbf{R_T = R_O (1+\alpha T)}$$

With a reference resistance $R_0$ of 2kΩ and a temperature coefficient $\alpha$=0.002/℃, we calculated the required resistance for the boundary conditions:

- For 0°C: The resistance was set to 2kΩ , representing the balanced state of the bridge.

- For 100°C: The resistance was increased to 2.45kΩ to simulate the maximum thermal input.

Important thing to mention is that, for this project the wheatstone bridge will be used in unbalanced state as only the unbalanced state induces voltage differential which is needed for this circuit to operate and return results.

## Designing the Amplifier:

From our previous calculations, we have derived an equation for RF and Rf = Rf1 = Rf2. Which is,

$$\boldsymbol{RF = \frac{2}{9}Rf}$$

Using PSpice to simulate our circuit we used various values of RF and Rf to marginalize our output from 0V to 5V.

| Trial 1 | |
|---|---|
| Rf = 1.5kΩ | RF = 333.33Ω |
| Found $V_0$ range | |
| At 0°C, $V_0$ = 0V | At 100°C, $V_0$ = 4.19V |

| Trial 2 | |
|---|---|
| Rf = 10kΩ | RF = 2.22kΩ |
| Found $V_0$ range | |
| At 0°C, $V_0$ = 0V | At 100°C, $V_0$ = 2.97V |

| Trial 3 | |
|---|---|
| Rf = 1.2kΩ | RF = 266.66Ω |
| Found $V_0$ range | |
| At 0°C, $V_0$ = 0V | At 100°C, $V_0$ = 4.62V |

| Trial 4 | |
|---|---|
| Rf = 1kΩ | RF = 222.22Ω |
| Found $V_0$ range | |
| At 0°C, $V_0$ = 0V | At 100°C, $V_0$ = 5.00V |

In our 4th trial, we found that the combination of Rf1 = Rf2 = Rf = 1kΩ and

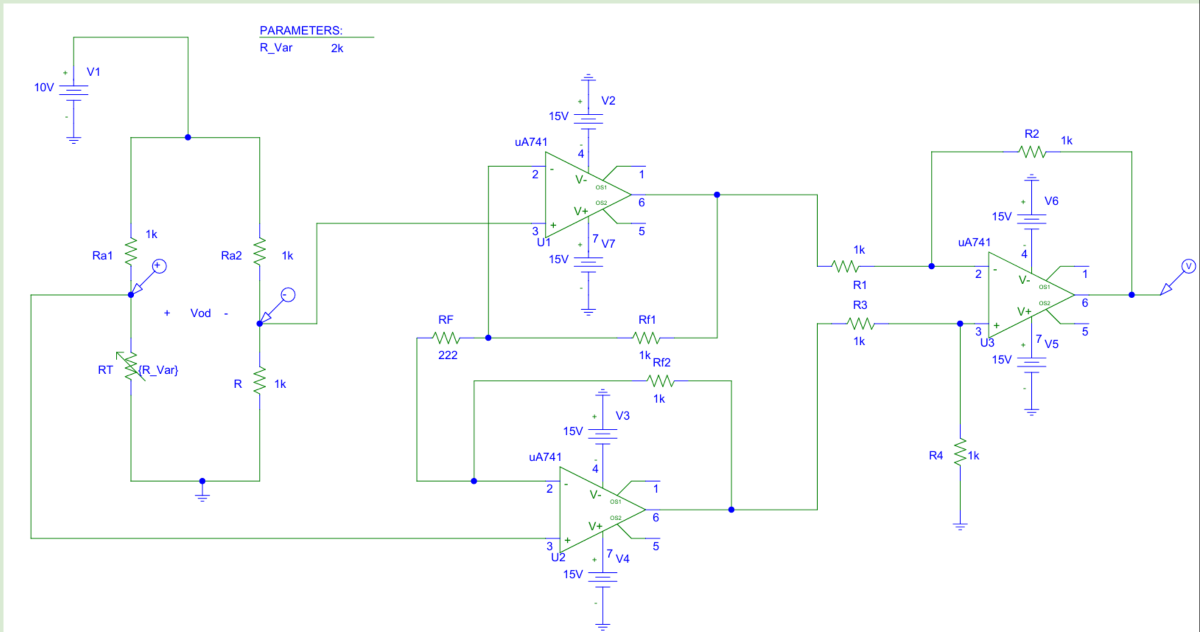
RF = 222.22Ω yielded the required output where $\boldsymbol{v_{od}}$ varies from 0V to 0.5V linearly while $\mathbf{V_0}$ varies from 0V to 5V.

This highlights that the gain is correct and as our variable resistor varies from 2kΩ to 2.45kΩ simulating the temperature change from 0°C to 100°C we will find corresponding values for $\boldsymbol{v_{od}}$ and $\mathbf{V_0}$.

| $\mathbf{R_T}$ | **T** | $\boldsymbol{v_{od}}$ | $\mathbf{V_0}$ |
|---|---|---|---|
| 2.000kΩ | 0°C | 0V | 0V |
| 2.040kΩ | 10°C | 0.05V | 0.5V |
| 2.082kΩ | 20°C | 0.10V | 1V |
| 2.124kΩ | 30°C | 0.15V | 1.5V |
| 2.167kΩ | 40°C | 0.20V | 2.0V |
| 2.21kΩ | 50°C | 0.25V | 2.5V |
| 2.256kΩ | 60°C | 0.30V | 3.0V |
| 2.301kΩ | 70°C | 0.35V | 3.5V |
| 2.348kΩ | 80°C | 0.40V | 4.0V |
| 2.396kΩ | 90°C | 0.45V | 4.5V |
| 2.445kΩ | 100°C | 0.5V | 5.0V |

Here, every value corresponds to each other clearly and our design is complete. To verify the circuit's operation we must simulate it in PSpice and also implement it in a breadboard to justify its operation.

# Experimental Verification of the Designed Circuit

<table>
<tr>
<td><u>Software Setup</u></td>
<td>
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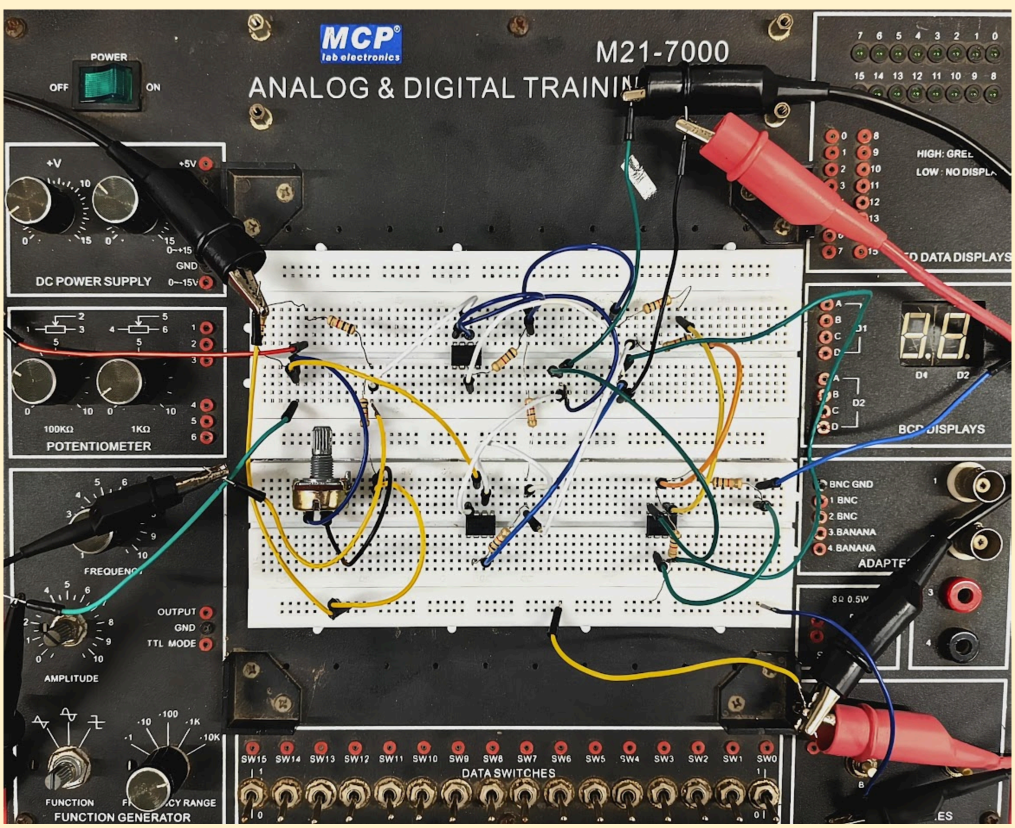
<tr>
<td><u>Hardware Setup</u></td>
<td>
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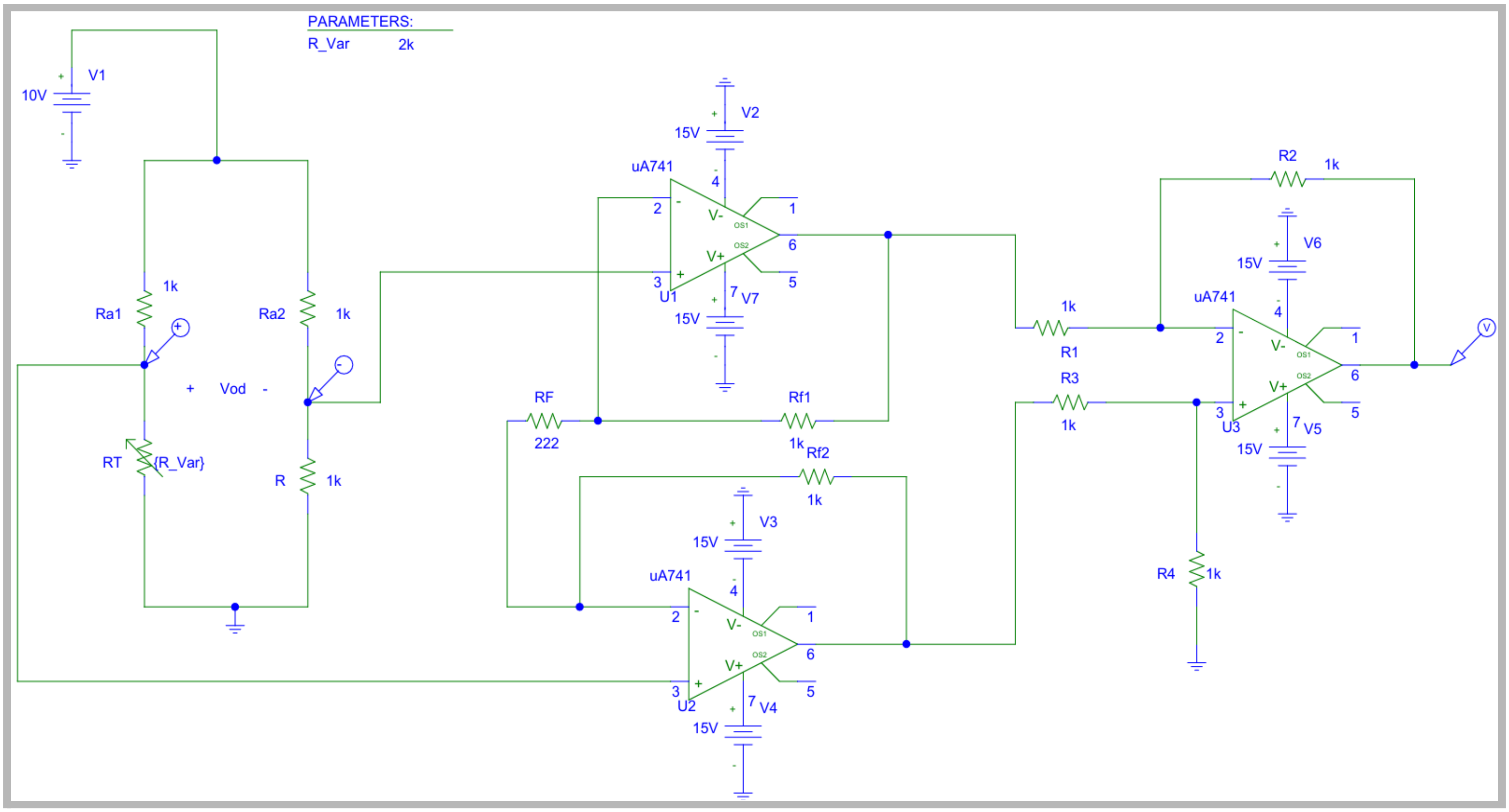
## Software Verification:




For our Software verification, we will use this circuit:

(i) <u>Setup:</u>

To simulate temperature variation from 0°C to 100°C, a variable resistor is used to represent the change in resistance caused by temperature. The parameter {R_Var} is defined and set in DC Sweep mode, allowing it to vary from 2 kΩ to 2.45 kΩ based on previous calculations.

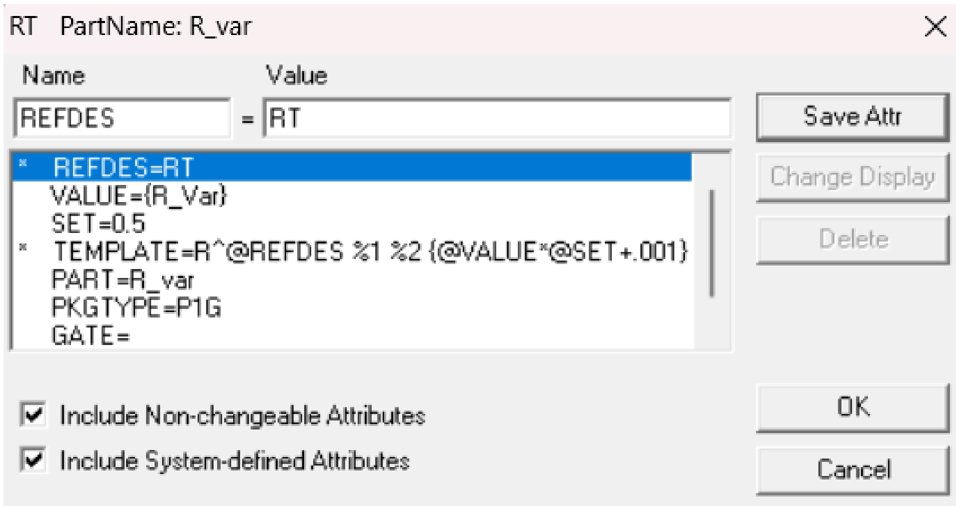



Fig:- {R_Var} Setup

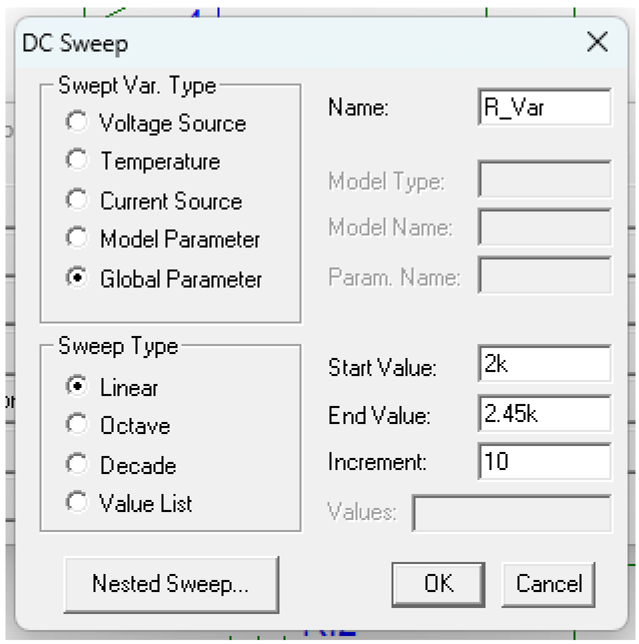



Fig:- DC Sweep Setup

After configuring the setup, the simulation is executed and the DC sweep result is selected to obtain the output graph. This graph shows the variation of output voltage with respect to the changing resistance. Finally, the results are observed to verify whether the circuit behaves as expected and produces a proper increase in voltage with increasing temperature.

(ii) Output Graphs

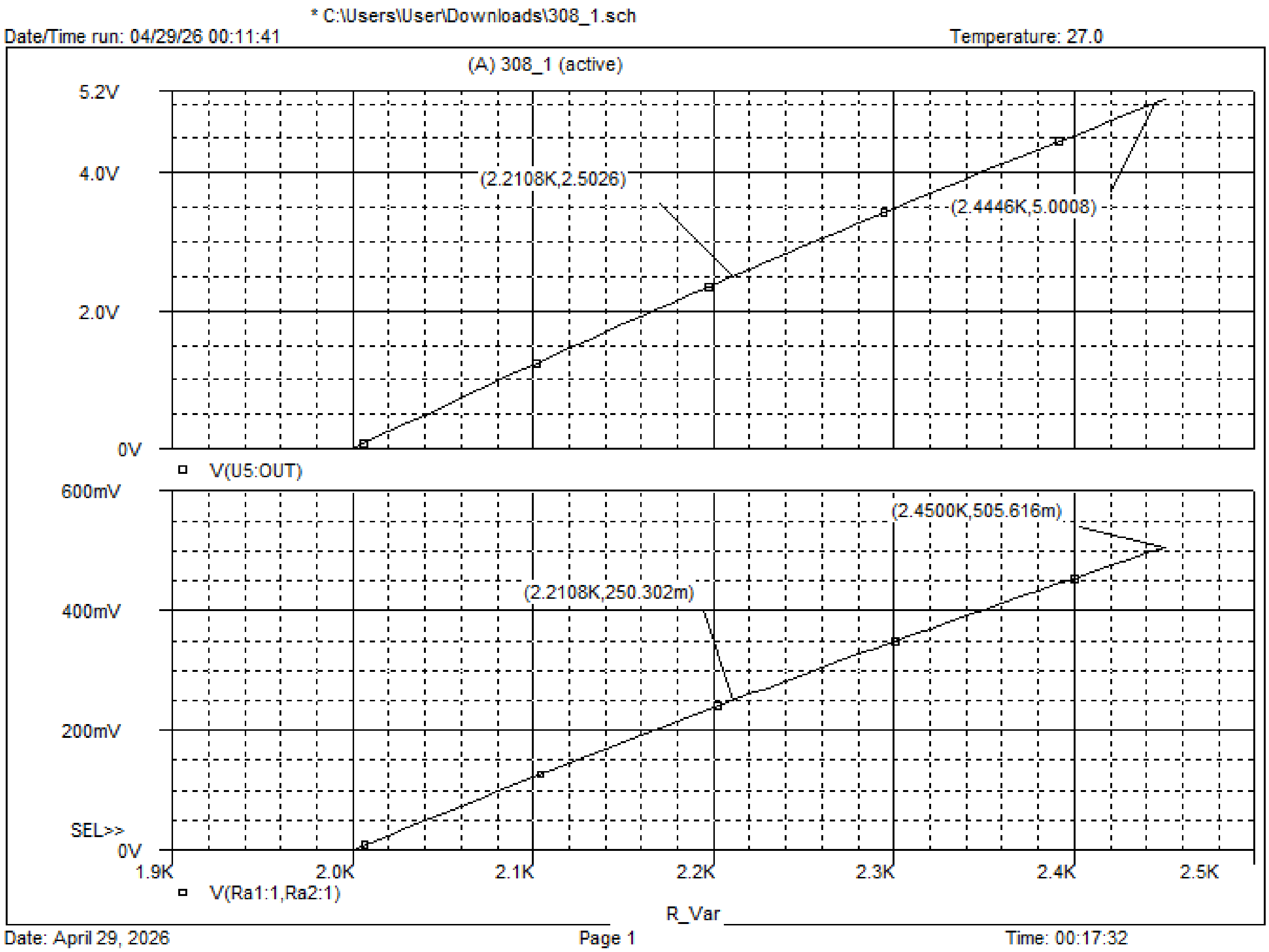
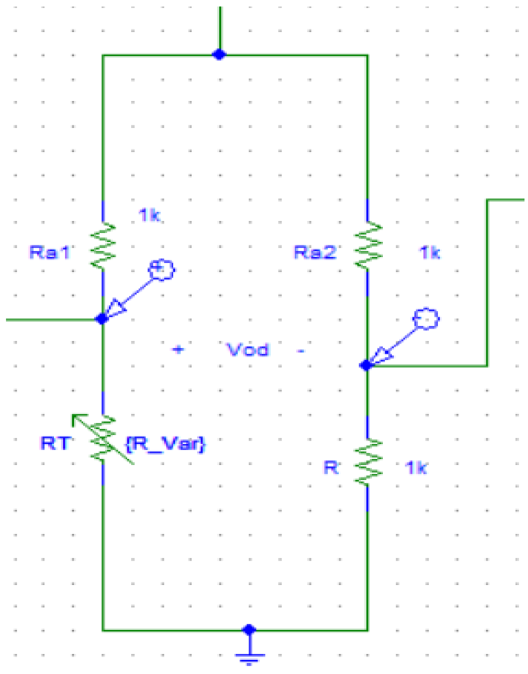



Fig:- The top graph is $V_o$ and the bottom graph is $V_{od}$

It shows that the voltage output is increasing accordingly to the voltage input of this circuit. For $V_{od}$ = 0.25 V (50℃), $V_o$ is showing 2.5V in output. And for $V_{od}$= 0.5V (100℃), $V_o$ is showing 5.00V in output. We have extended the x axis to accommodate the output voltage.

(iii) <u>Calculation Review</u>

There are mainly 3 parts in this circuit. Firstly, the wheatstone bridge, where the variable resistor $R_T$ is placed. Instead of a temperature sensor, we used this variable resistor to imitate the temperature reaction on this circuit.

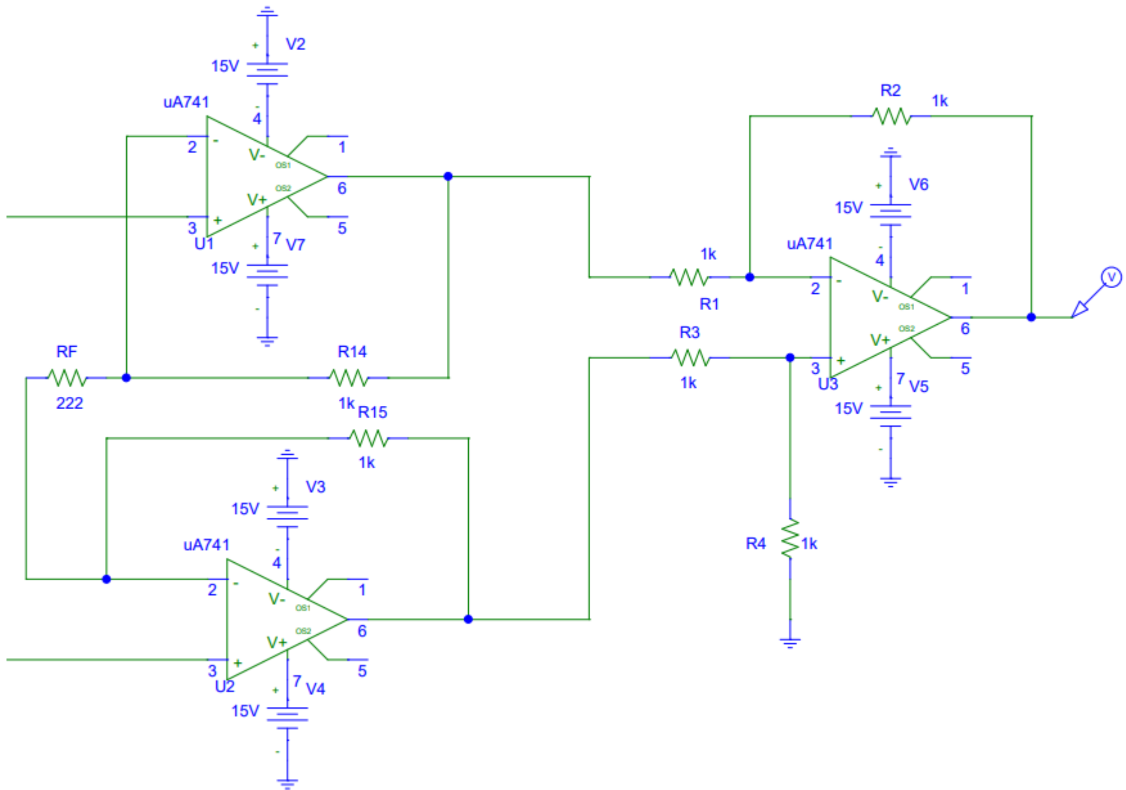



It is mainly the variable resistor on the basis of temperature value .

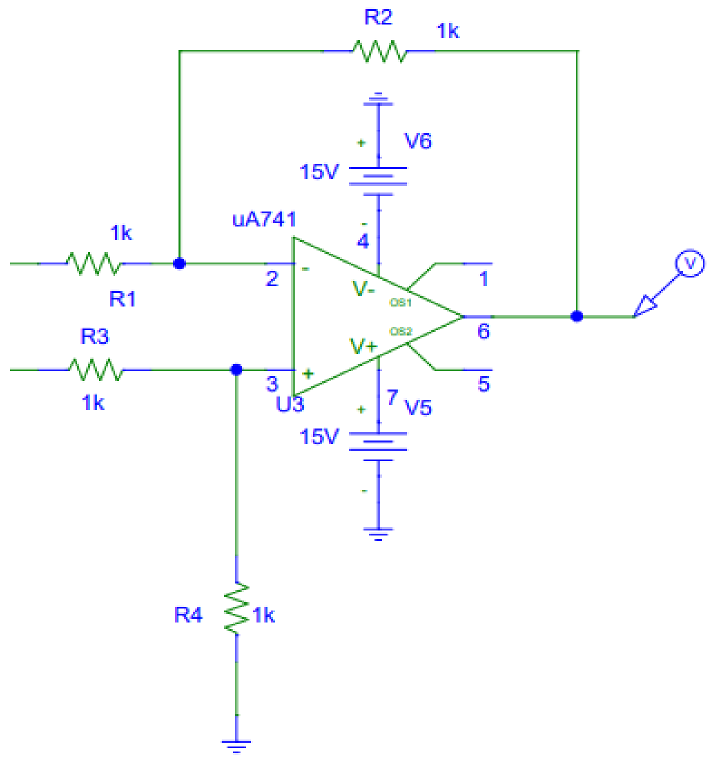
$$R_T = R_0 ( 1 + \alpha T )$$

We calculated that, at 0℃ $R_T$ = 2k and at 100℃ RT = 2.45k.

The voltage differential marker is placed between Ra1 and RT for the (+) marker and Ra2 and R for the (-) marker which showed us the graph for Vod .




The left half of the instrumentation amplifier, we have to check with the relation of Rf and RF. The equation for them was $9Rf = 2RF$




The differential amplifier gives us the output Vo . It shows us all the corresponding output graphs to vary almost up to 5V. In here we have to make sure the ratio is $\frac{R2}{R1} = \frac{R4}{R3} = 1$

## Hardware Verification


Fig:- The highest value of potentiometer

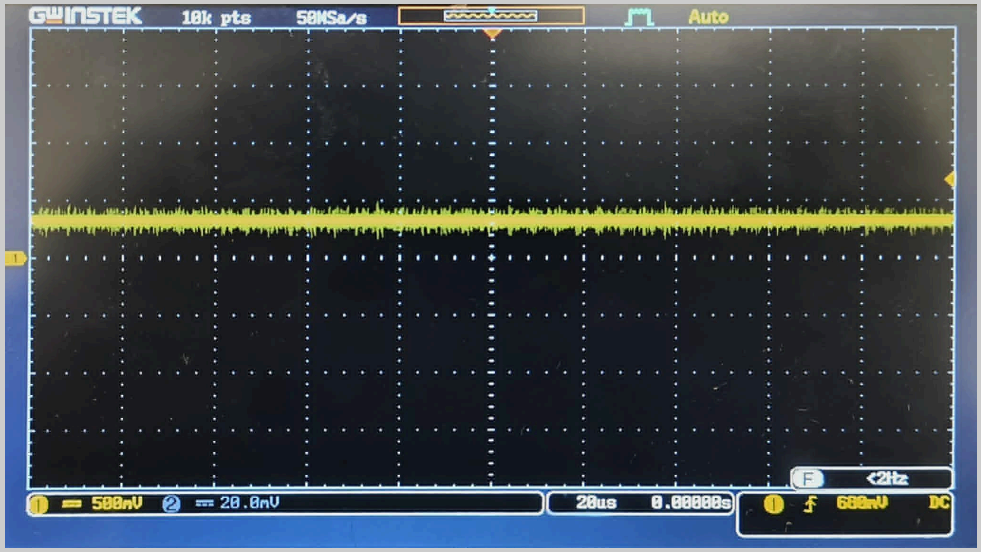
To vary T, we used a potentiometer to vary the resistance from 2kΩ to 2.45kΩ. Due to the highly sensitive nature of the potentiometer knob, the final value was varied to **2.47kΩ** for the maximum desirable value of $v_{od}$ and

**Op-amps power supply (+15V & -15V)**

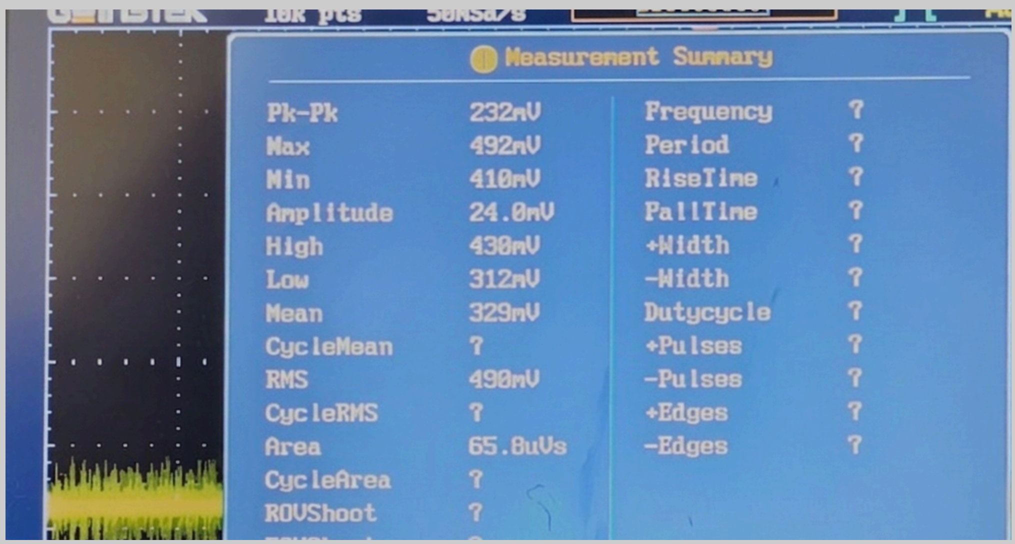

**Reference Voltage for Wheatstone bridge( $V_{ref}$ = 10V )**

**For Vod=492mV**

Input Waveform, $\boldsymbol{v_{od}}$ at,
$R_T$ = 2.47kΩ,
Simulating T = 100℃



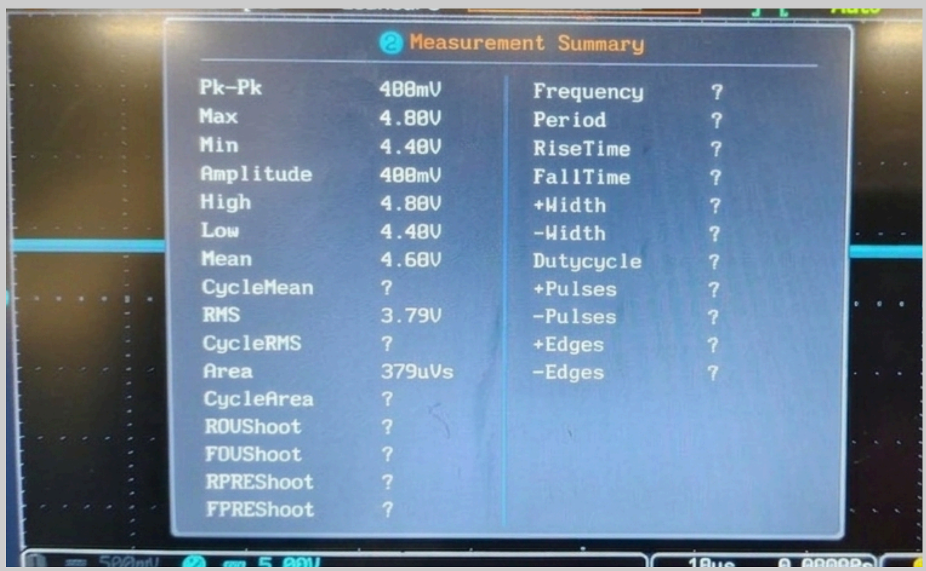

Value of $\boldsymbol{v_{od}}$ at
$R_T$ = 2.47kΩ,
SImulating T = 100℃



**For Vo=4.80V**

Output Waveform $\mathbf{V_0}$ at,
$R_T$ = 2.47kΩ,
Simulating T = 100℃


Value of $\mathbf{V_0}$ at,
$R_T$ = 2.47kΩ,
Simulating T = 100℃



| T (℃) | $V_{od}$ (V) | $V_o$ (V) Theoretical | $V_o$ (V) Simulation | $V_o$ (V) Hardware |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 |
| 50 | 0.25 | 2.5 | 2.50 | 2.5-2.7 |
| 100 | 0.5 | 5.0 | 5.0 | 4.80 |

## Error:-

**Gain error:-**

For ideal case,

At highest temperature $V_{od}$=0.5V & Amplifier Gain, $A_v$=10

So we get output,

$V_0=A_v*V_{od}=10*0.5=5V$

But in our hardware time we have

For ,$V_{od}$=0.5V , $V_0$=4.80V

Amplifier gain will be, Av= 4.80/0.5=9.6

It means that our gain dropped from 10 to 9.6

So, Gain Error= {(10-9.6)/10}*100%

= 4%

**Resistor Mismatch in Amplifier:**

we assume that,

$Rf=1k\Omega$, $RF=222\Omega$

$A_v=\{1+(2*R_f)\}/R_F\}$

$= \{1+(2*1)\}/0.222\}$

$=10$

But In hardware we have ,

$R_{f1} =0.997k\Omega$

$R_{f2} =0.994k\Omega$

$R_F=218\Omega$

$R_1= 0.996k\Omega$

$R_2= 0.987k\Omega$

$R_3=0.992k\Omega$

$R_4= 0.994k\Omega$

When we vary the potentiometer,it is very hard to fix this value in 2.45kΩ. So we fix that problem by pick a closest value,so we pick 2.47kΩ

$Av=4.8/0.492= 9.756$

Hardware Error $=\{(5-4.8)/5\}*100\%$

$=4\%$

| Temperature (℃) | Theoretical Value (V) | Hardware Value (V) | Error(%) |
|---|---|---|---|
| 10 | 0.5 | 0.48 | 4% |
| 20 | 1.0 | 0.96 | 4% |
| 30 | 1.5 | 1.44 | 4% |
| 40 | 2.0 | 1.92 | 4% |
| 50 | 2.5 | 2.4 | 4% |
| 60 | 3.0 | 2.88 | 4% |
| 70 | 3.5 | 3.36 | 4% |
| 80 | 4.0 | 3.84 | 4% |
| 90 | 4.5 | 4.32 | 4% |
| 100 | 5.0 | 4.8 | 4% |

The table shows a steady, linear relationship between temperature and voltage. The hardware values are consistently 4% lower than the theoretical values across the entire range, indicating a systematic gain error rather than random noise. This steady deviation is likely caused by small queries in resistor tolerances or amplifier settings. Since the error is constant and predictable, the system remains highly linear and can be easily corrected with simple calibration.

**Software Simulation Error:**

$V_{od}$=505mV=0.505V
$V_o$=5.00V

Error=$(\frac{5-5.00}{5}) * 100\%$
=0%

Gain Av=$V_o$ / $V_{od}$
=5/0.505
= 9.9
Gain Error = $(\frac{10-9.9}{10})$*100%
= 1%

## Design Requirements & Specifications

**Equipments:**

1. Op amp(UA741)
2. Resistor (1kΩ with 5% tolerance, 220Ω)
3. Potentiometer (0-10kΩ)
4. Jumper wire
5. Breadboard
6. DC power supply
7. PSpice Software

## Requirements

### System Requirements

- The system is required to measure temperature over a range of 0°C to 100°C with a clear and proportional electrical output.
- The output voltage should be 0V at 0°C and increase linearly up to 5V at 100°C.
- The circuit must ensure a linear relationship between temperature and output voltage for accurate interpretation.

### Signal Specifications

- The Wheatstone bridge should generate a small differential voltage that varies approximately from 0V to 0.5V.
- The overall system should maintain signal integrity during amplification.

### Power and Biasing Conditions

- A constant and stable DC reference voltage (10V) is required to operate the bridge circuit.
- The amplifier section should be powered using a dual supply to allow proper signal amplification.
- Voltage fluctuations and noise should be minimized to ensure reliable operation.

## Specifications

### Component Specifications

Operational Amplifier

- The op-amp should have high input impedance to avoid loading the Wheatstone bridge.
- It should have low offset voltage and stable gain for accurate amplification.

Resistors

- The selected values must match the calculated design to ensure proper operation.
- Resistors with low tolerance (preferably ±1%) are recommended to reduce error.

Temperature Sensing Element

- An RTD is considered as the sensing element due to its linear response with temperature.
- For testing and simulation, a variable resistor (potentiometer) is used instead of a real RTD.

### Performance Specifications

- Output varies linearly with temperature
- Small error present (4%)
- Output is stable and consistent
- System shows predictable behavior

Design Specifications

- Simple circuit structure
- Low-cost components
- Easy to test and adjust
- Easy to build on breadboard

## Comments and Conclusions

The designed digital thermometer system works as expected and shows a clear linear relationship between temperature and output voltage. Both simulation and hardware results follow the theoretical values closely which proves that the Wheatstone bridge and instrumentation amplifier were properly designed and implemented. Although a small error of around 4% was observed in the hardware output it remained consistent across all measurements indicating a simple issue such as resistor tolerance or minor component mismatch rather than random error. This means the system is still reliable and can be improved further by using more precise components. Overall, the project successfully demonstrates how temperature can be converted into a measurable voltage signal and it provides a good practical understanding of circuit design.